105.863574 lines to be studied

TIKLE'S ACADEMY

1) Line is parallel to both HP & VP.

2) Line is on both HP & VP.

3) Line is Perpendicular to HP & parallel to VP.

4) Line is Perpendicular to VP & parallel to HP.

5) Line is incline to HP & parallel to VP.

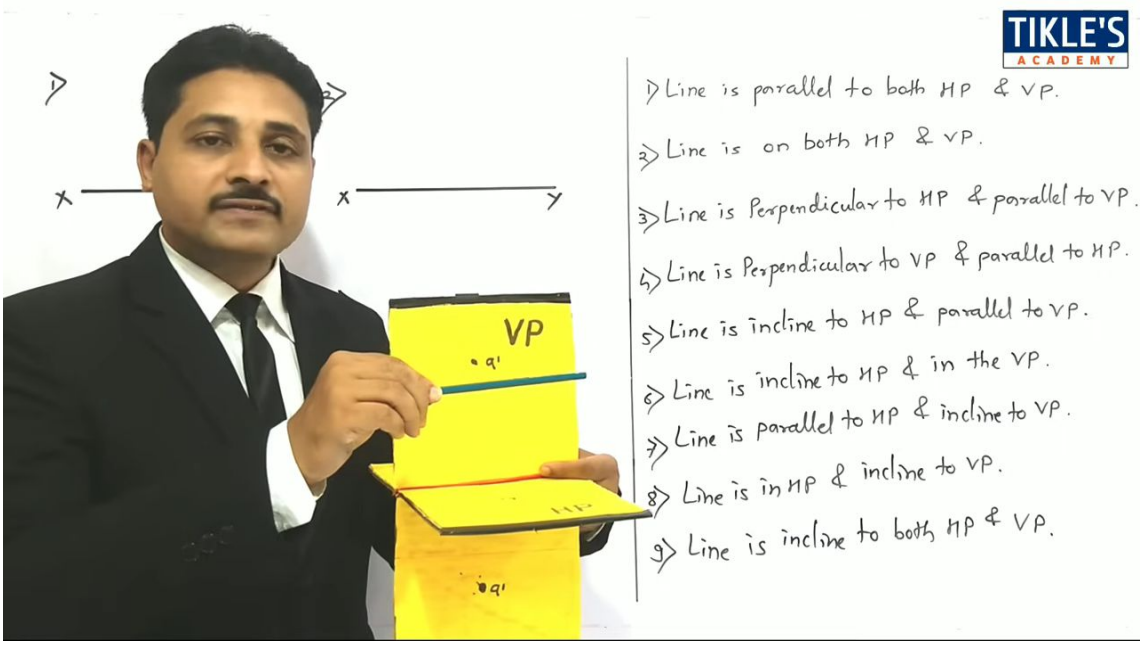
6) Line is incline to HP & in the VP.

7) Line is parallel to HP & incline to VP.

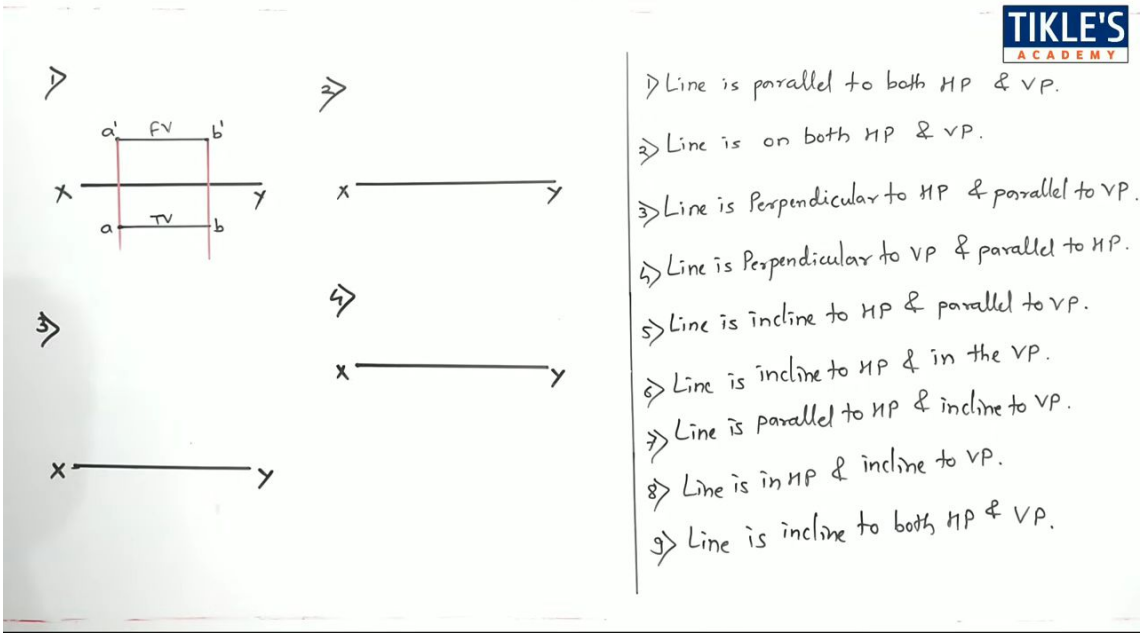
8) Line is in HP & incline to VP.

9) Line is incline to both HP & VP.

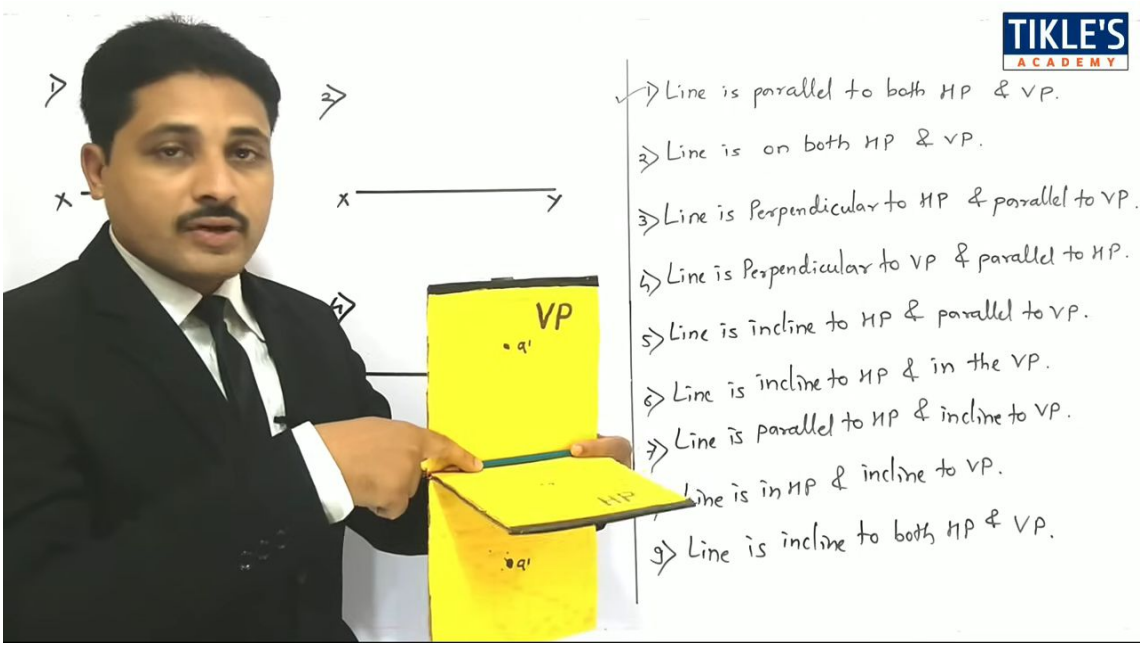
446.348991 line is parallel to both vp and hp



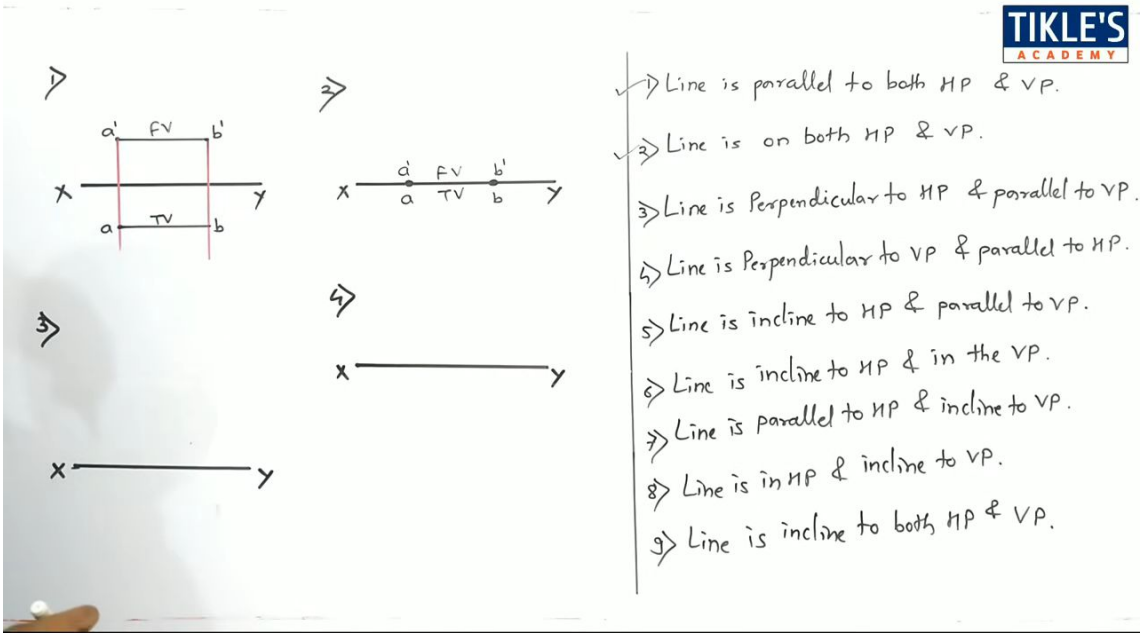
534.172109 line is parallel to both vp and hp



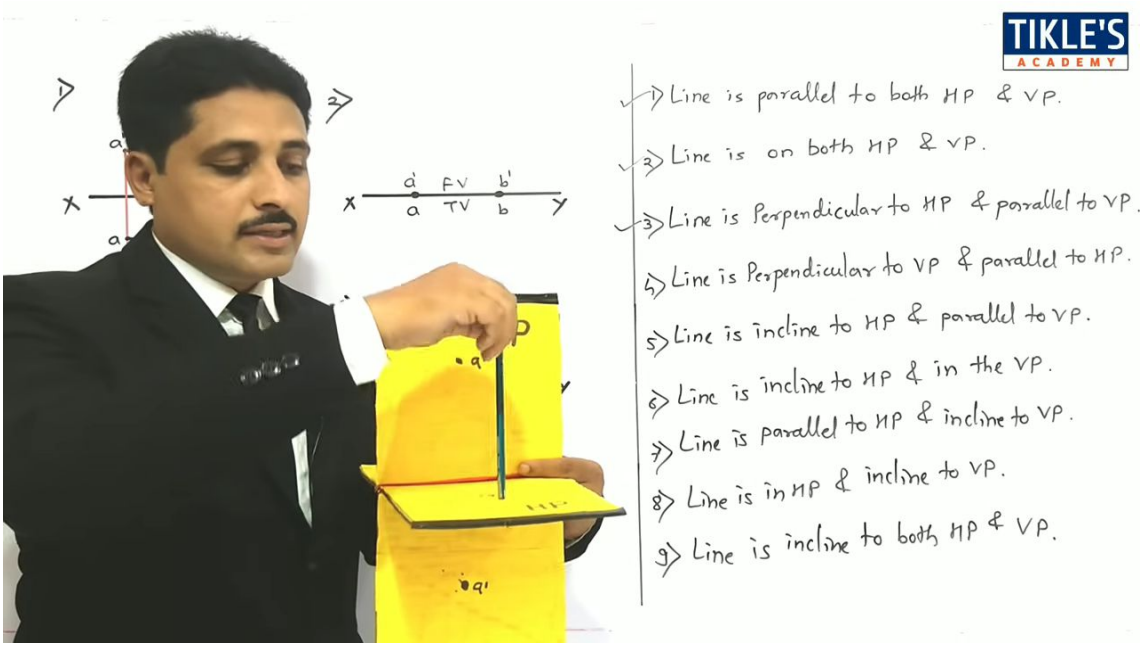
560.75251 line is on both hp and vp



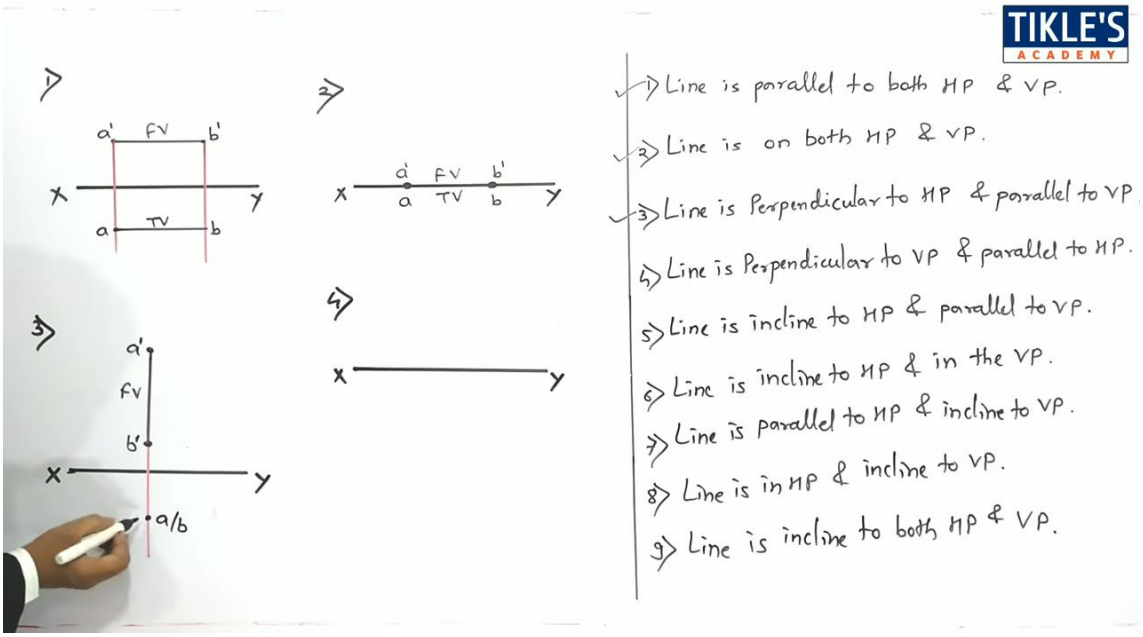
[613.034144](#) line is on both hp and vp



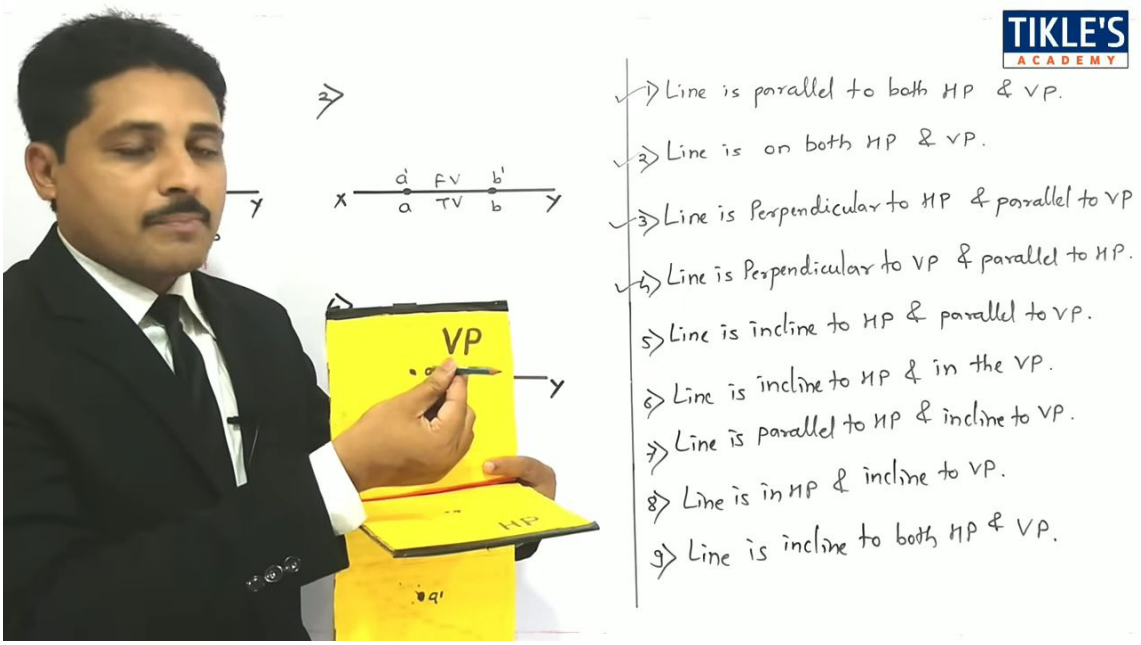
[651.88199](#) line is perpendicular to hp and parallel to vp



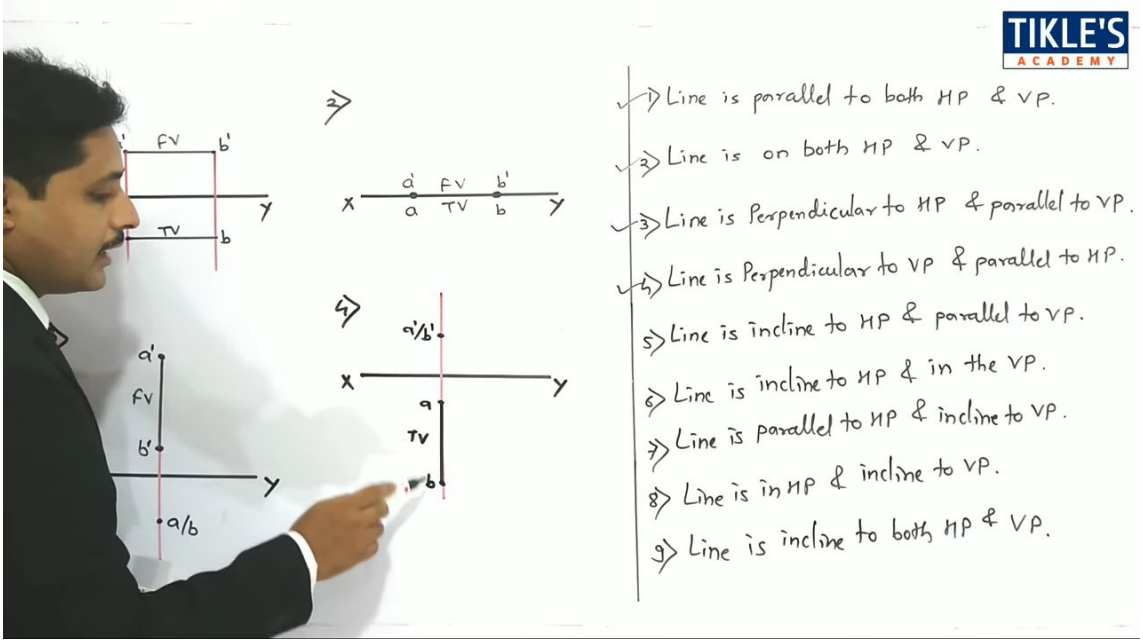
750.249024 line is perpendicular to hp and parallel to vp



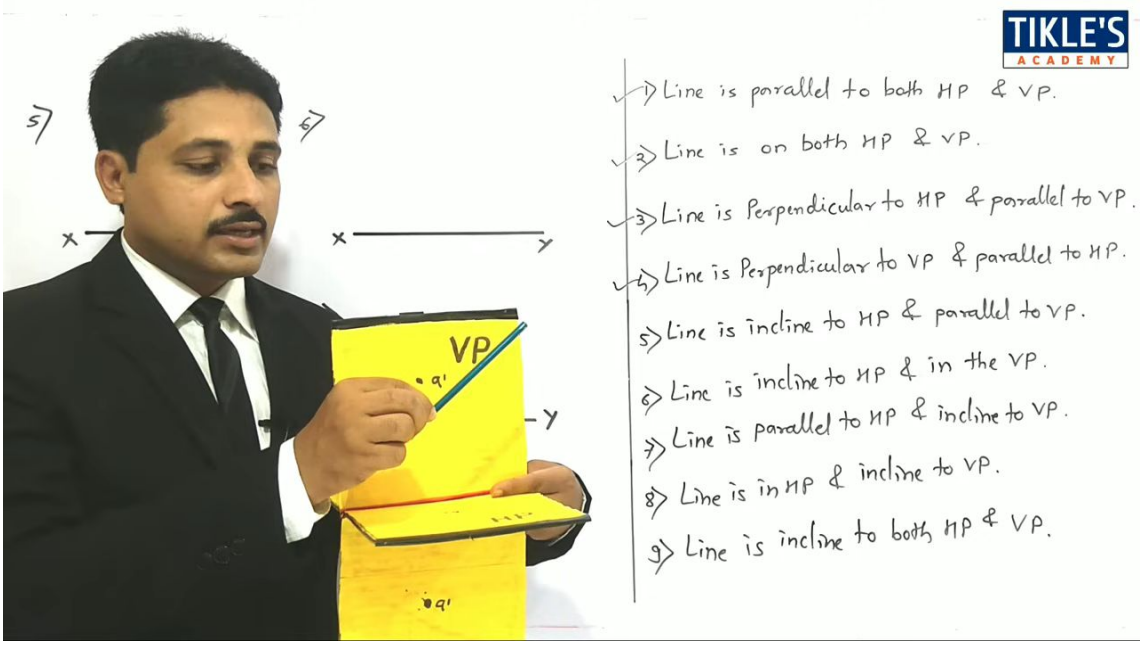
798.178655 line is perpendicular to vp and parallel to hp



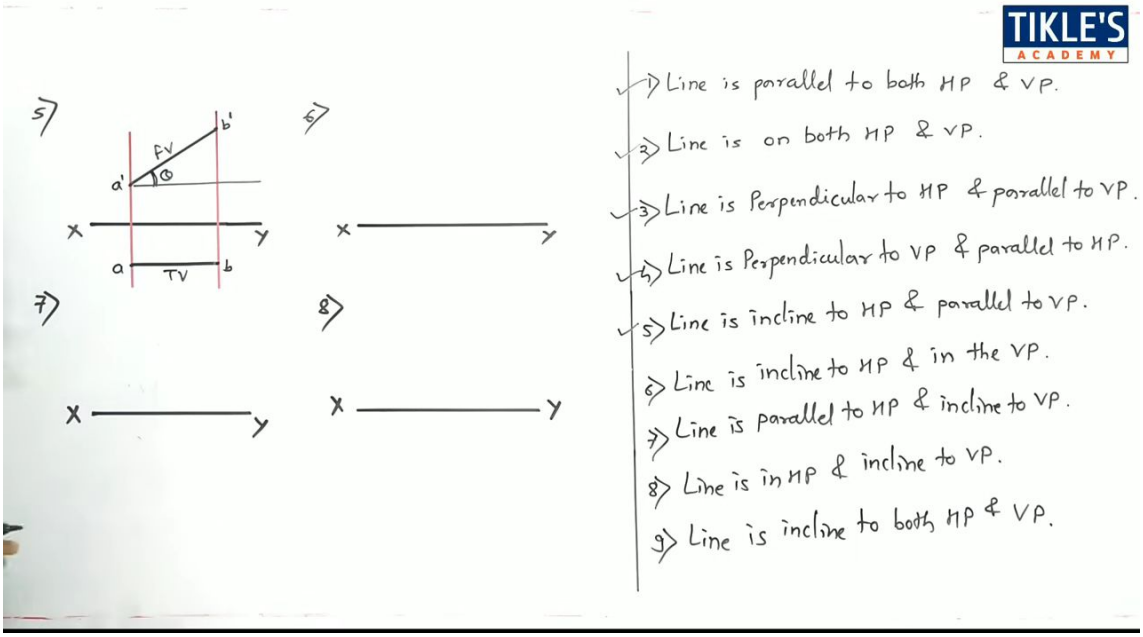
876.129332 line is perpendicular to vp and parallel to hp



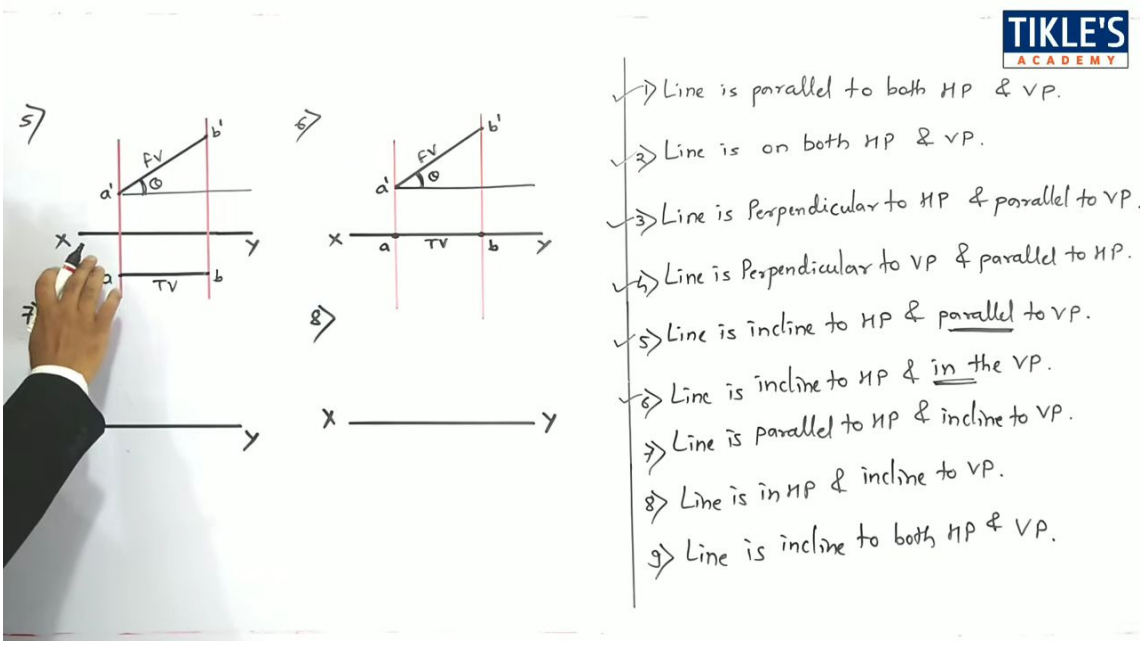
910.641373 line is inclined to hp and parallel to vp



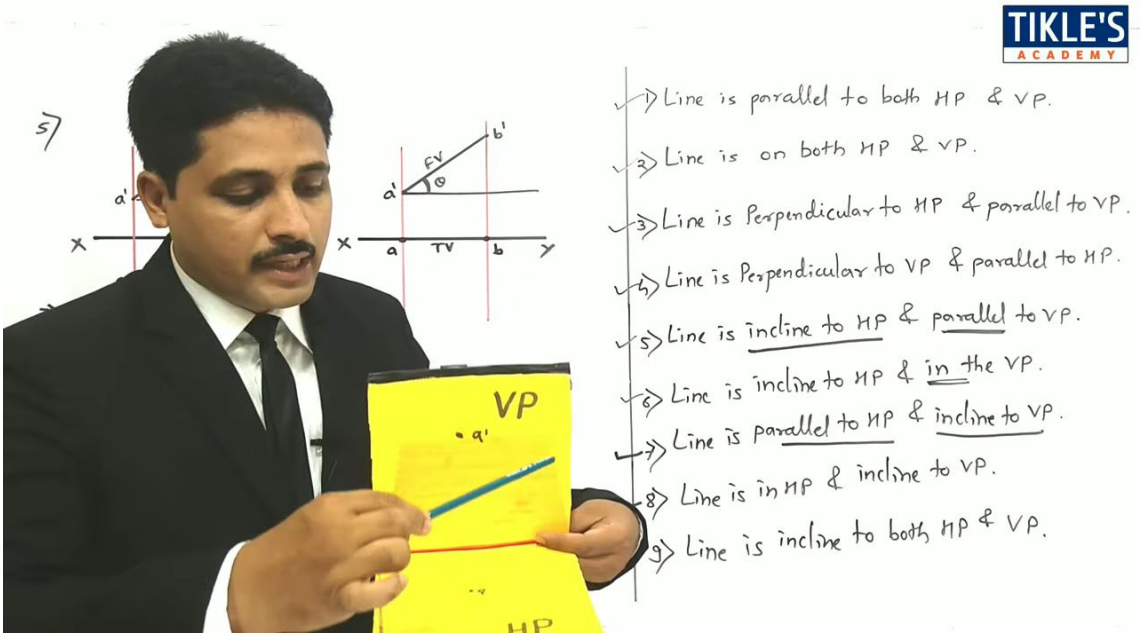
1043.953929 line is inclined to hp and parallel to vp



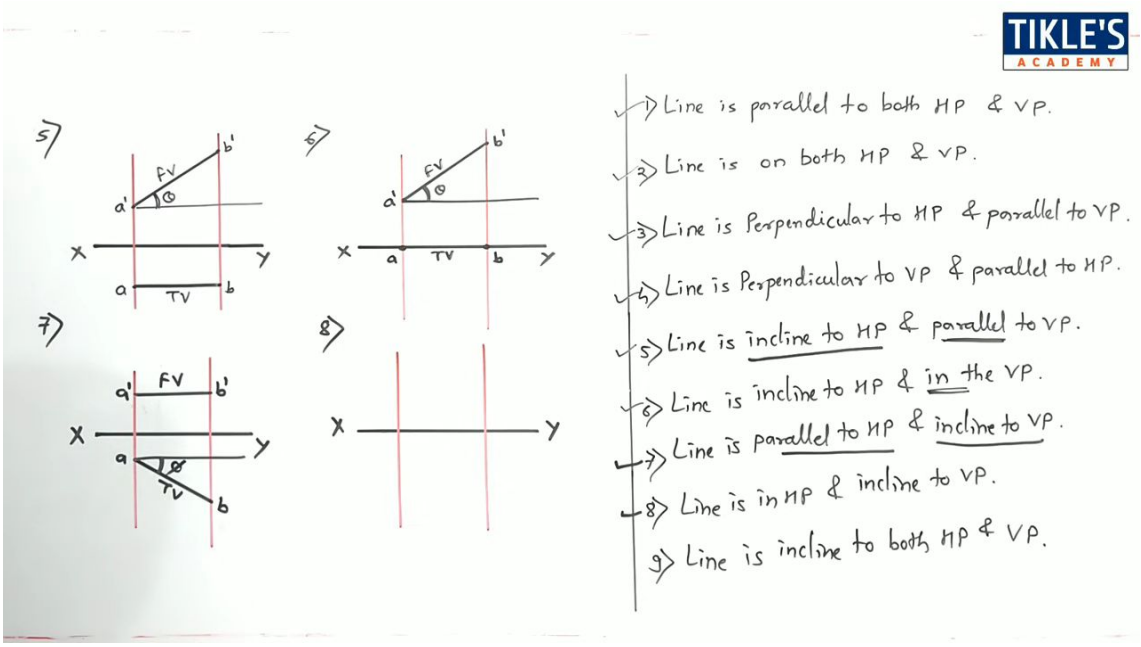
1138.645629 line is inclined to hp and in vp



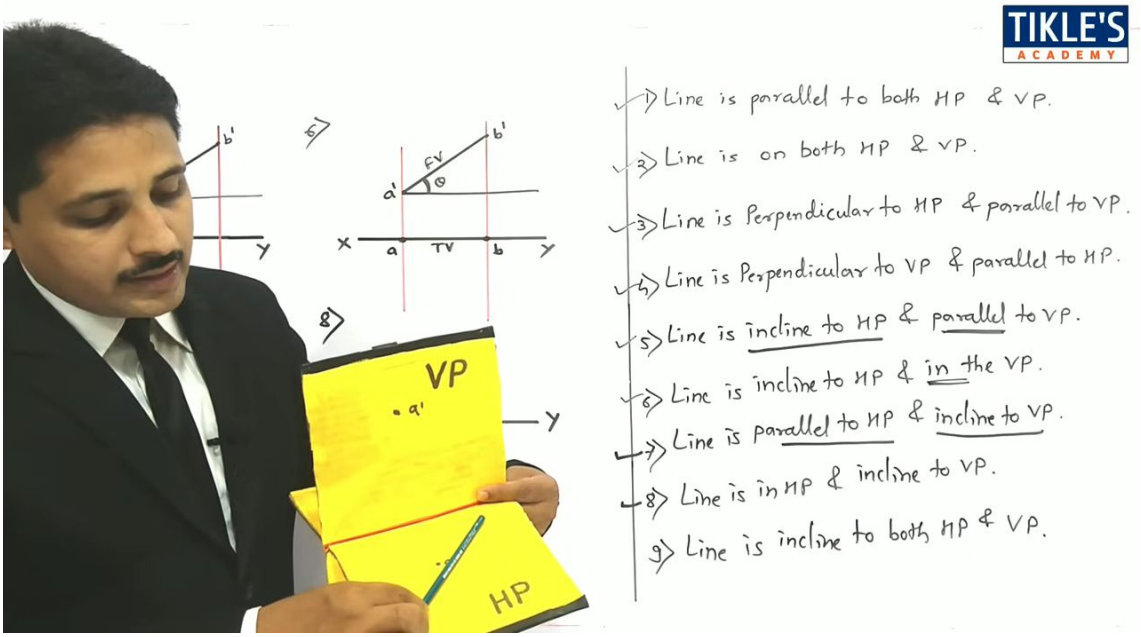
1209.381331 line is parallel to hp and inclined to vp



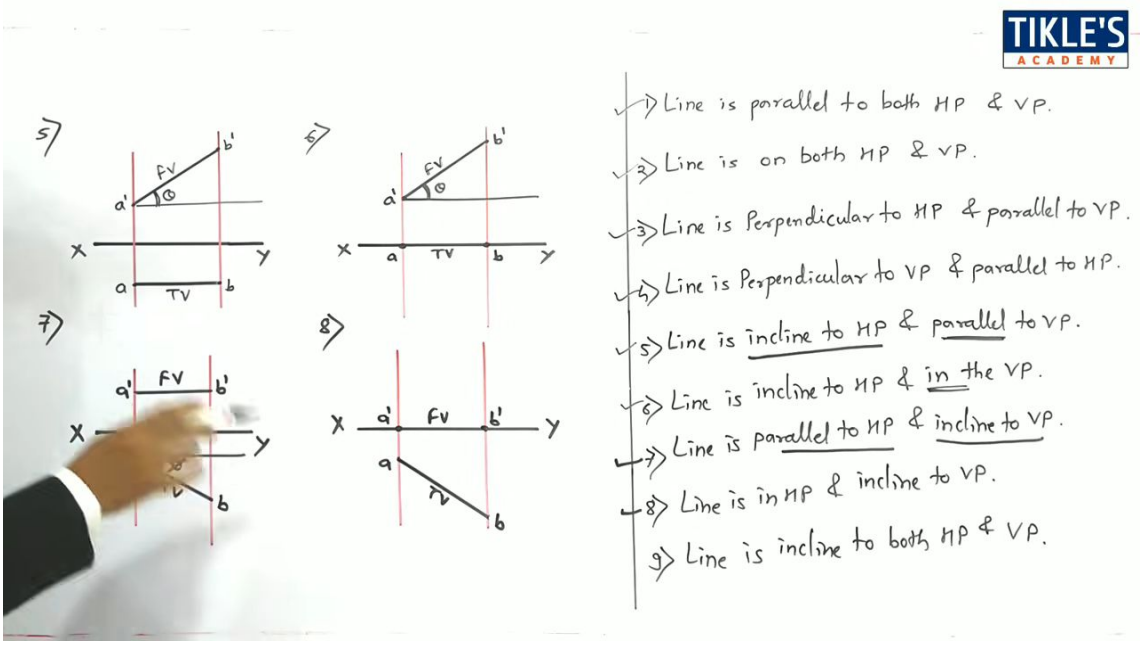
1353.253191 line is parallel to hp and inclined to vp



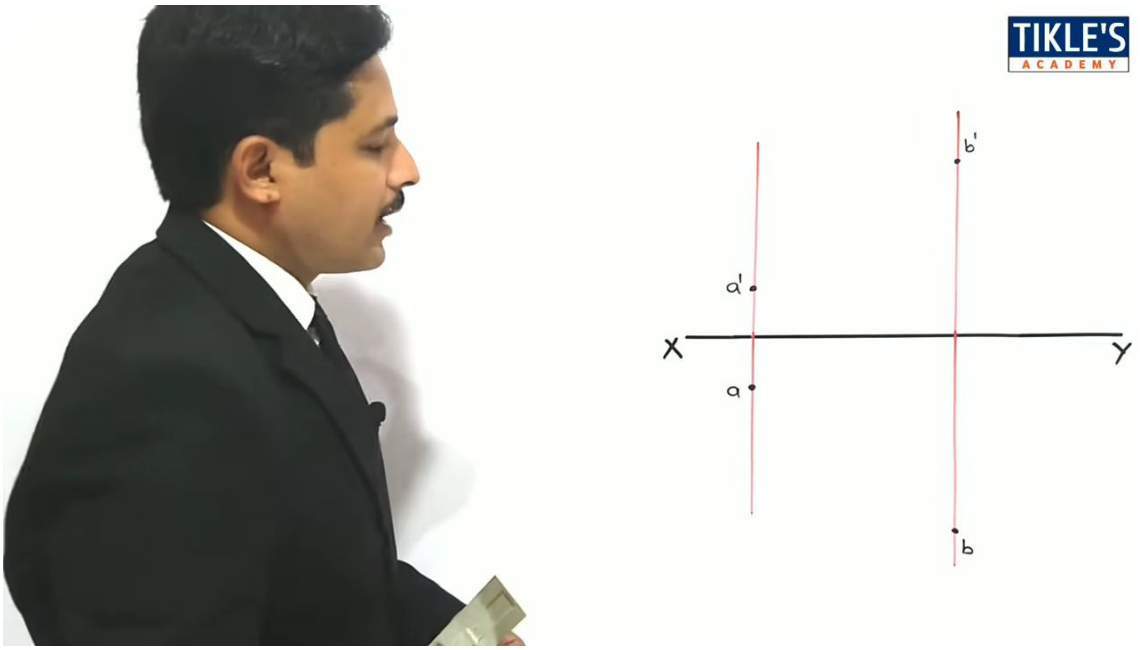
1265.885538 line is in hp and inclined to vp



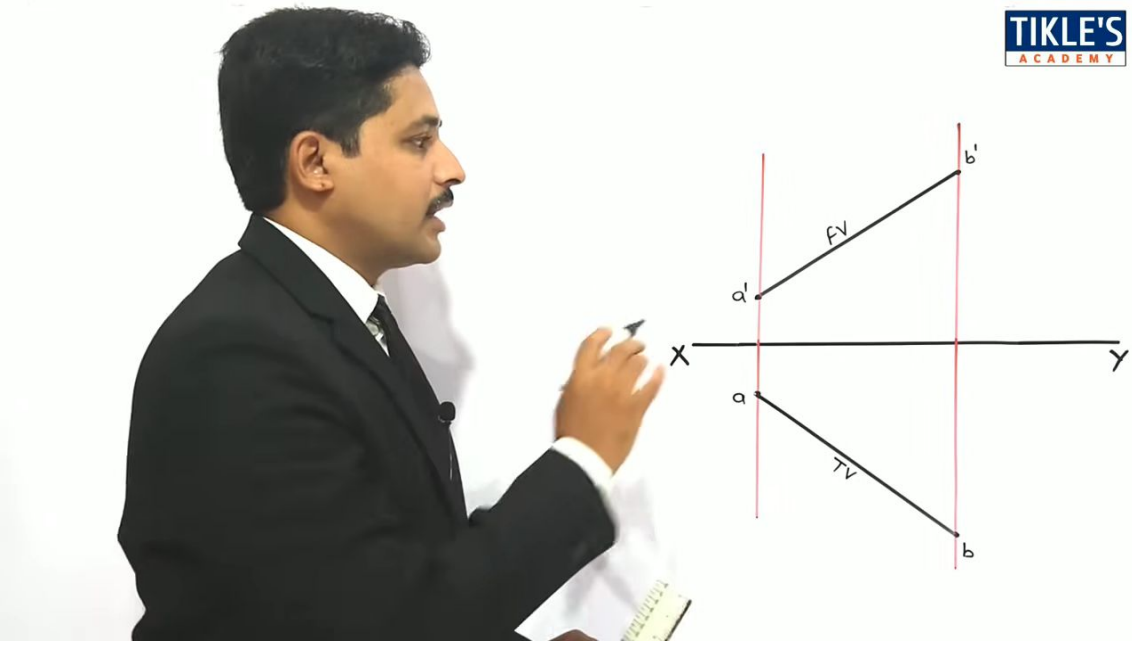
1407.899404 line is in hp and inclined to vp



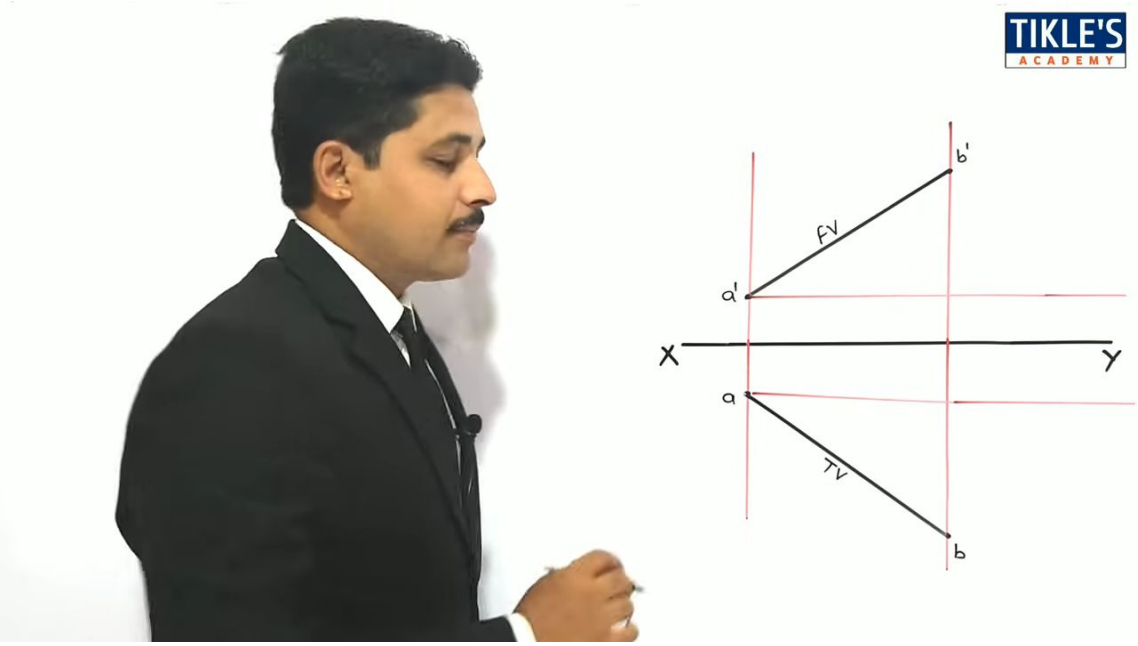
454.479447 we drew the projectors (red lines) to draw two points of the line. {Study is of 1st quadrant rn} The distance between these two red lines is called distance between the end projectors.



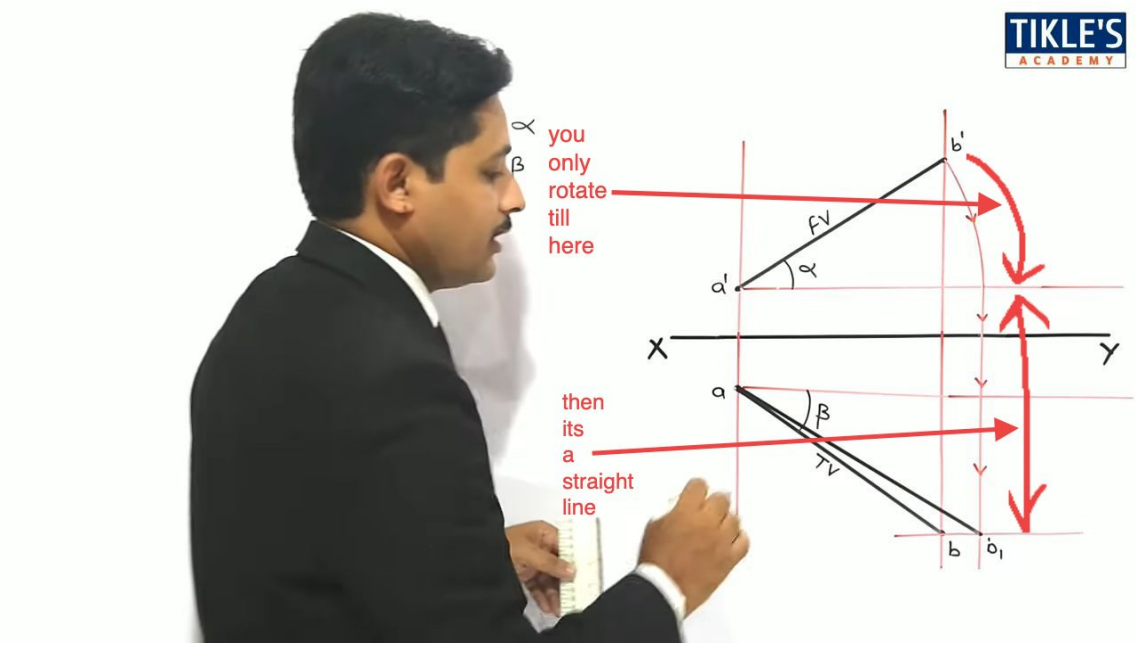
603.478925 joining both the fv points and tv points gives us the length of fv and tv.



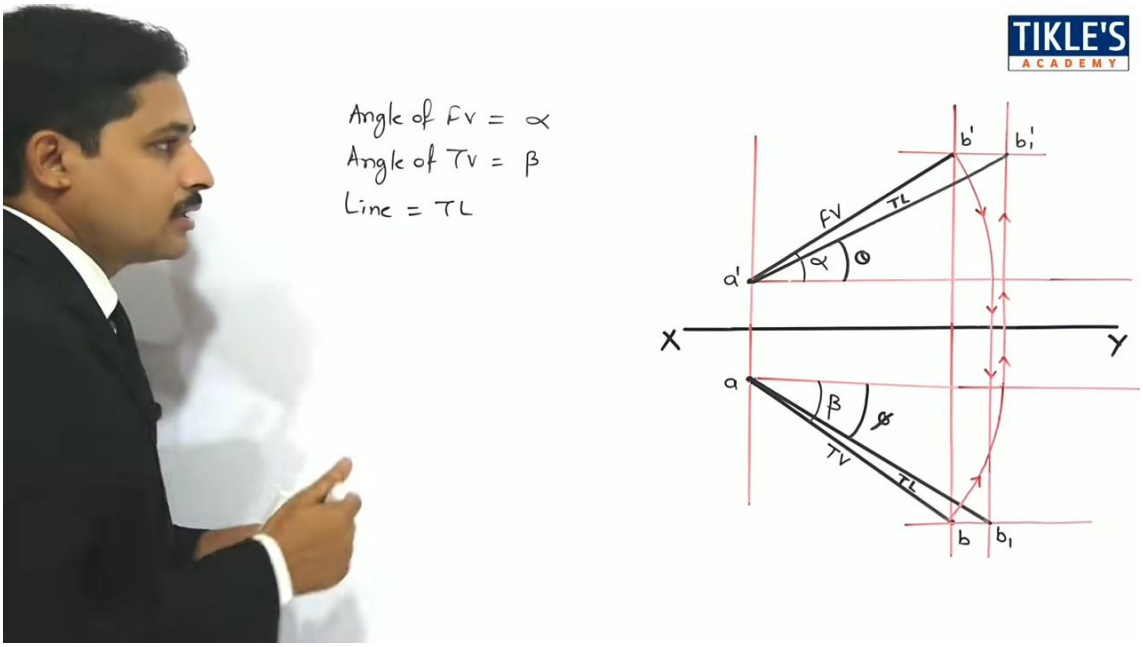
FV and TV's points are always in a vertical straight line Rotating and taking the vertical projection of FV gives the Front True Length. While the same for TV gives Top True Length. 721.227015 we draw two horizontal lines first.



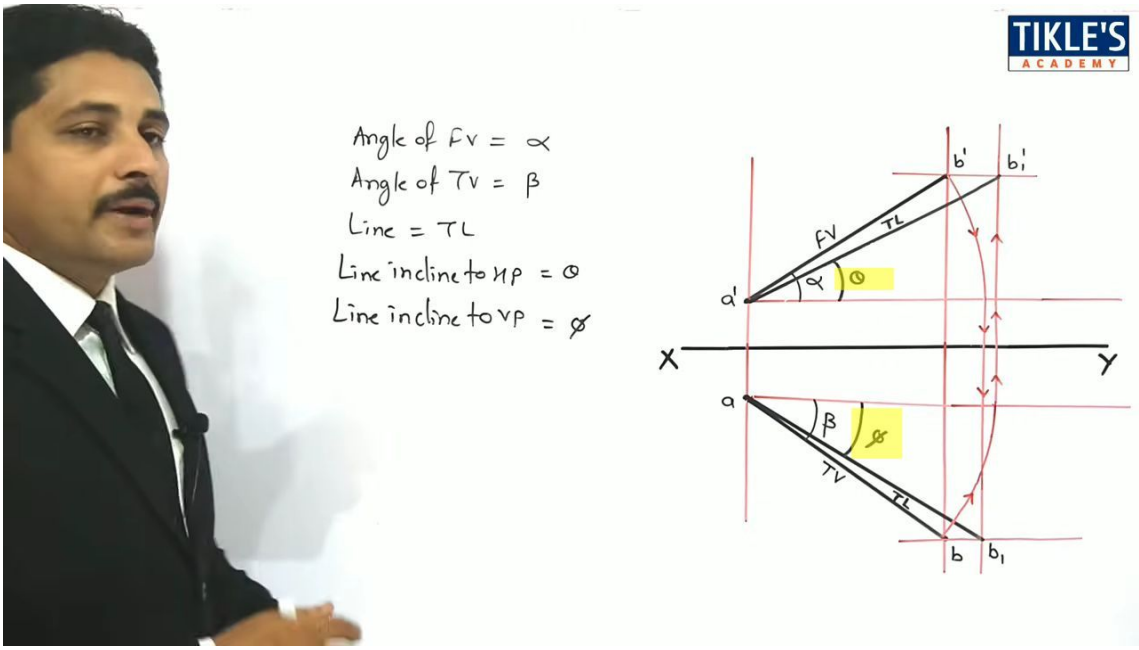
878.292761 then we project the point b' on the opposite side taking a' as the center. We also draw a perpendicular line to our end projector at b. This line's intersection with our rotated projection gives b1. Joining this with a gives us front true length.



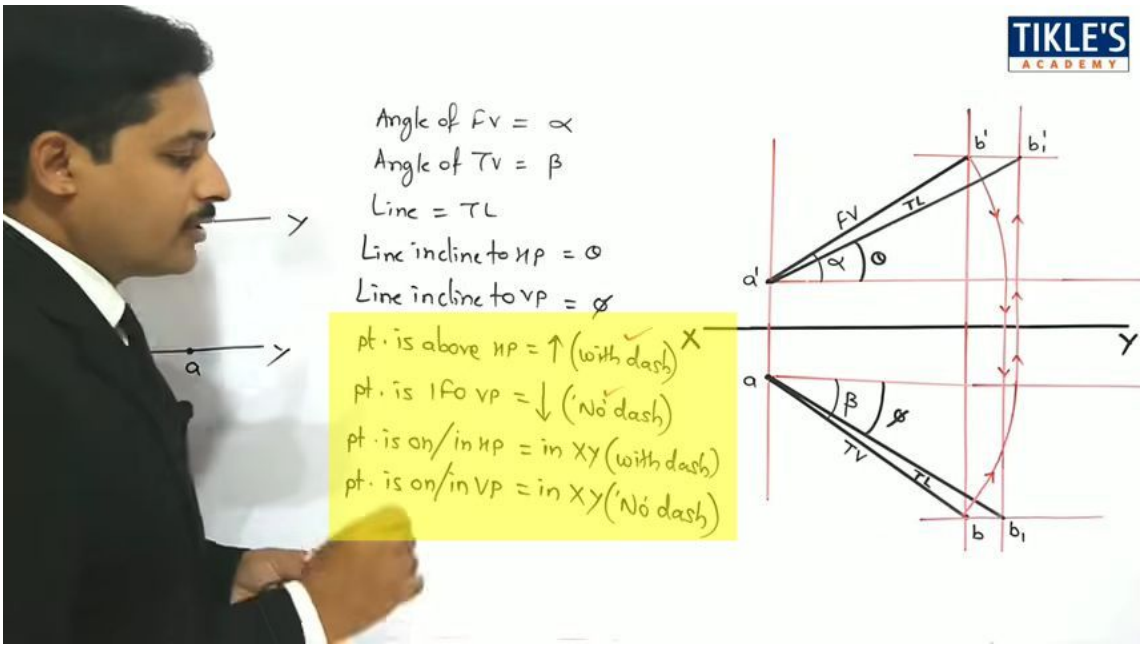
1041.336297 similarly we find the true length of top view



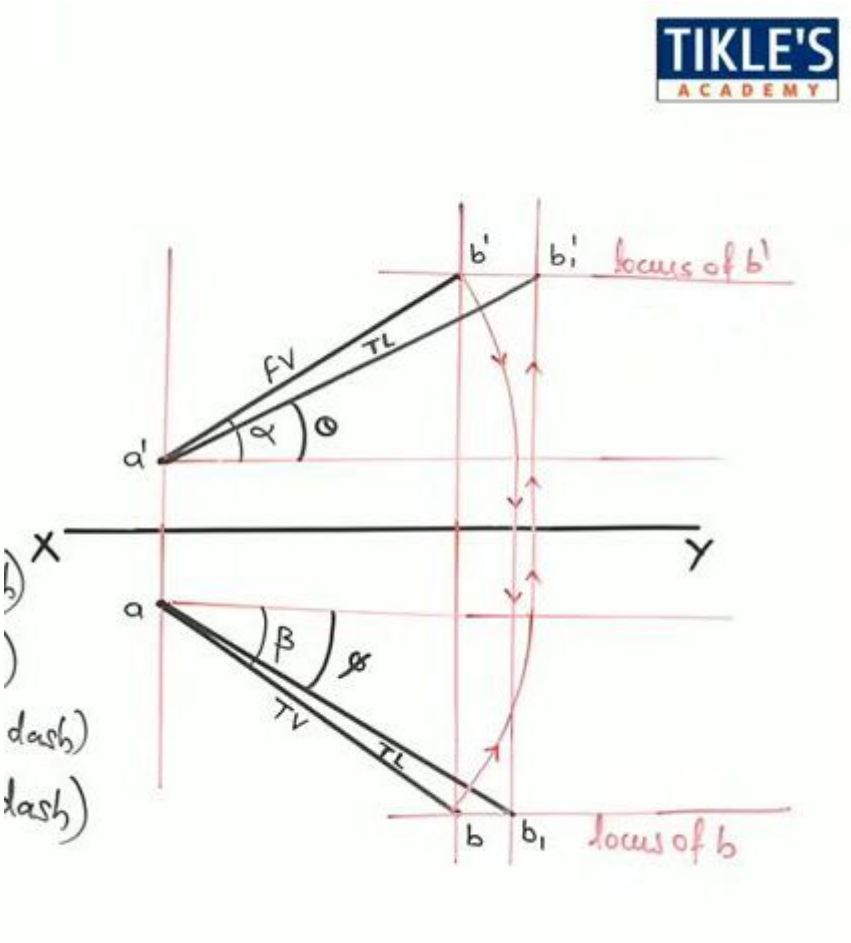
line inclined to hp/vp = angle between their true length and horizontal. 1127.254788



[1365.829316](#)

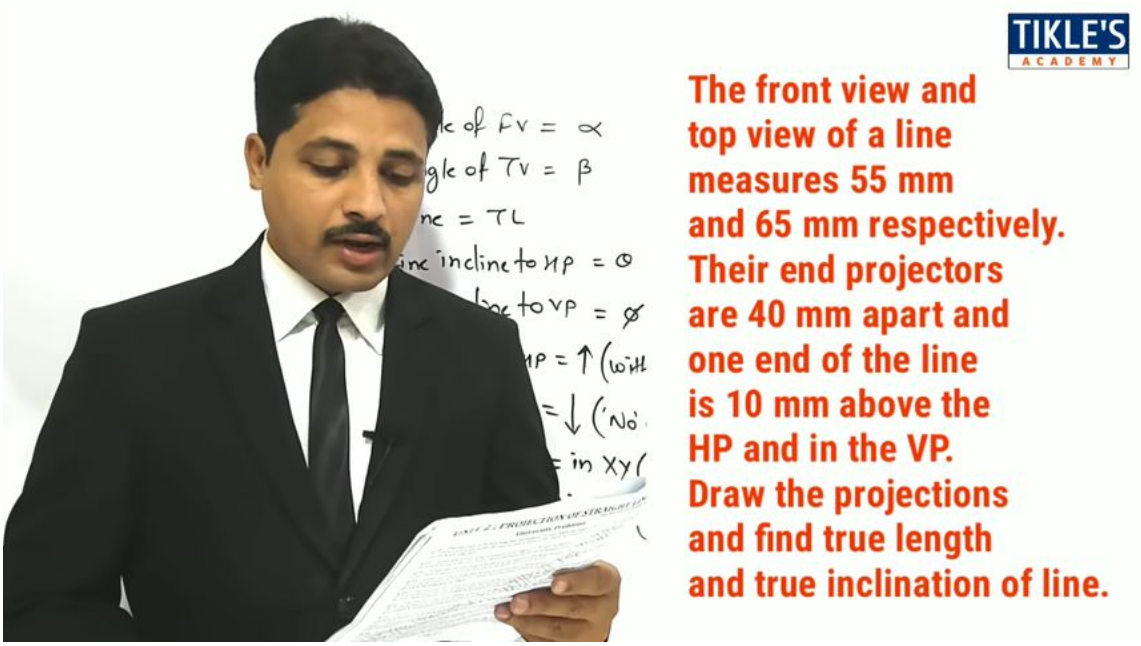


TV and top true length points as well as FV and front true length points are in a horizontal line/level called Locus.

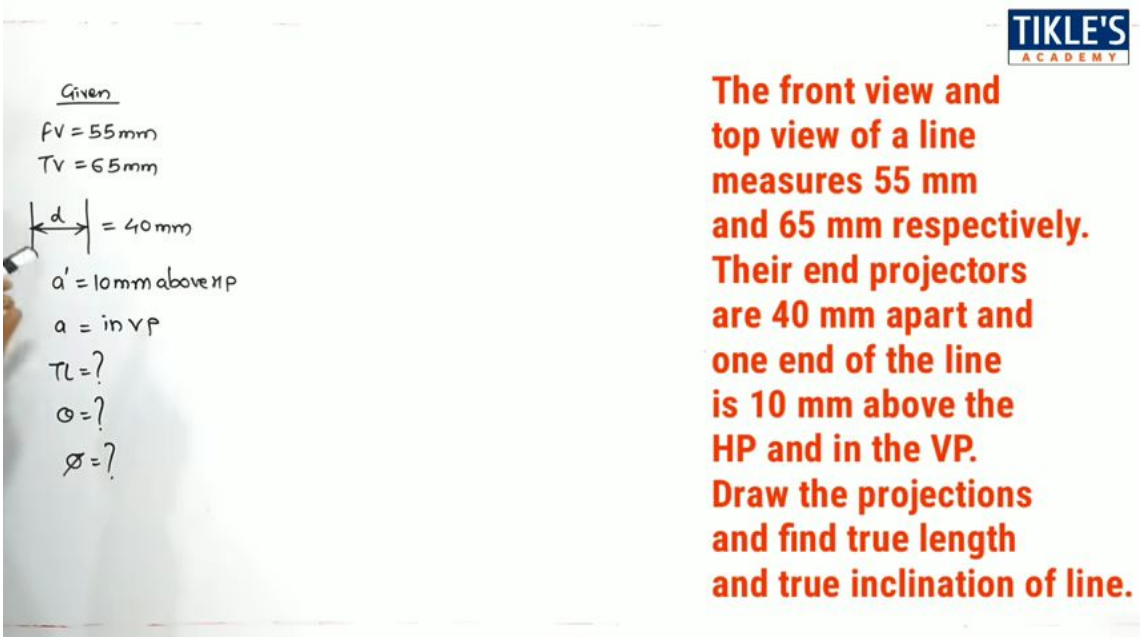


1509.444335 Also, the true length points are always right of the fv/tv points. Always. FV and TV's lengths are always lesser than the true lengths. True length points are not necessarily in a vertical straight line. But, length of both true lengths are always SAME.

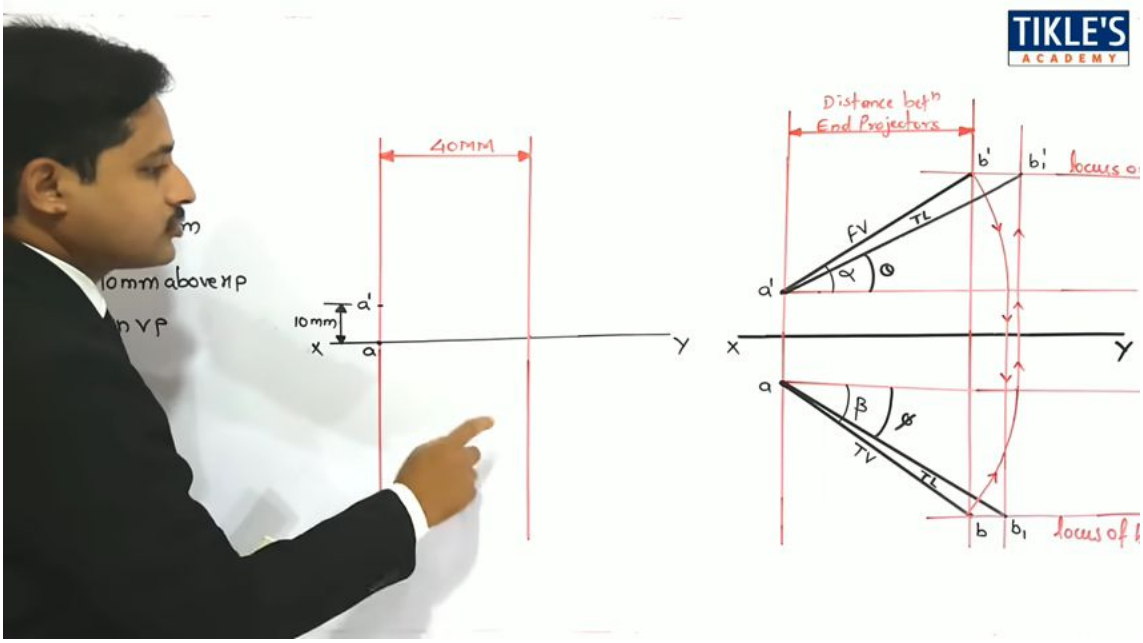
Q1. 93.984832



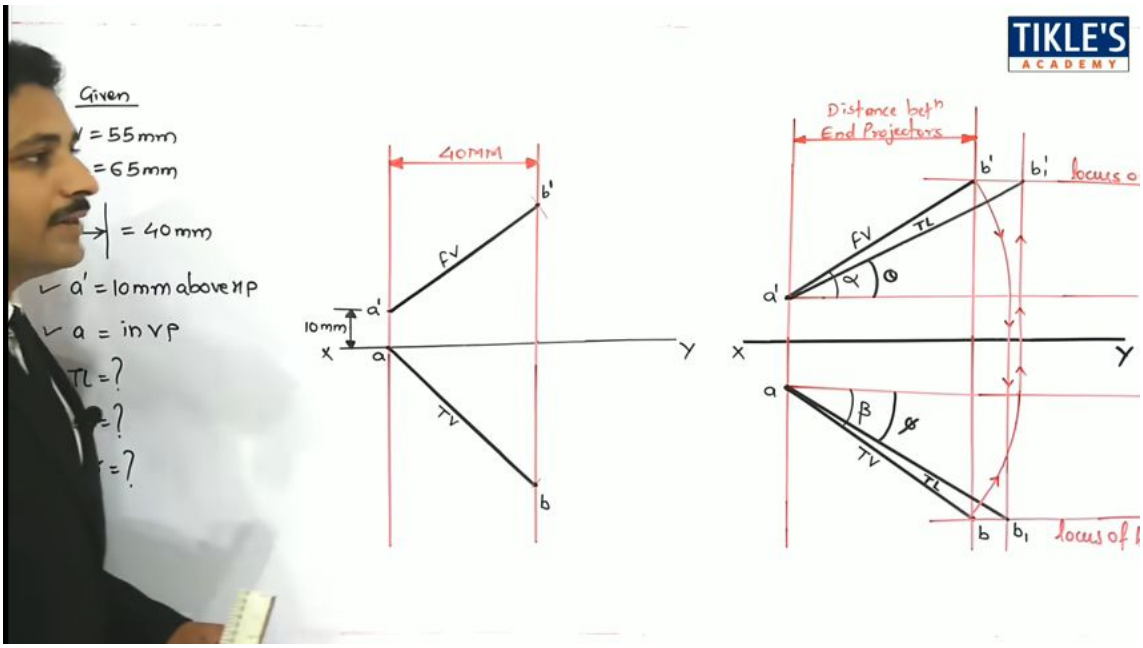
first write the given data. Each ques has minimum 5 data. 293.321765



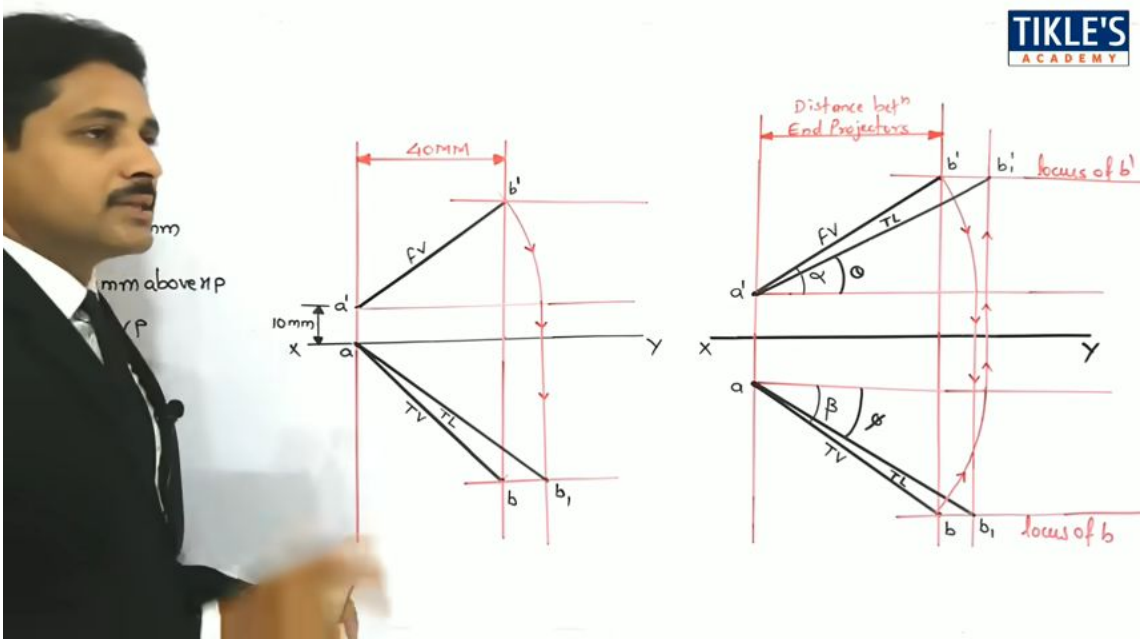
472.72846 first, we drew end projectors 40 mm apart, a and a'.



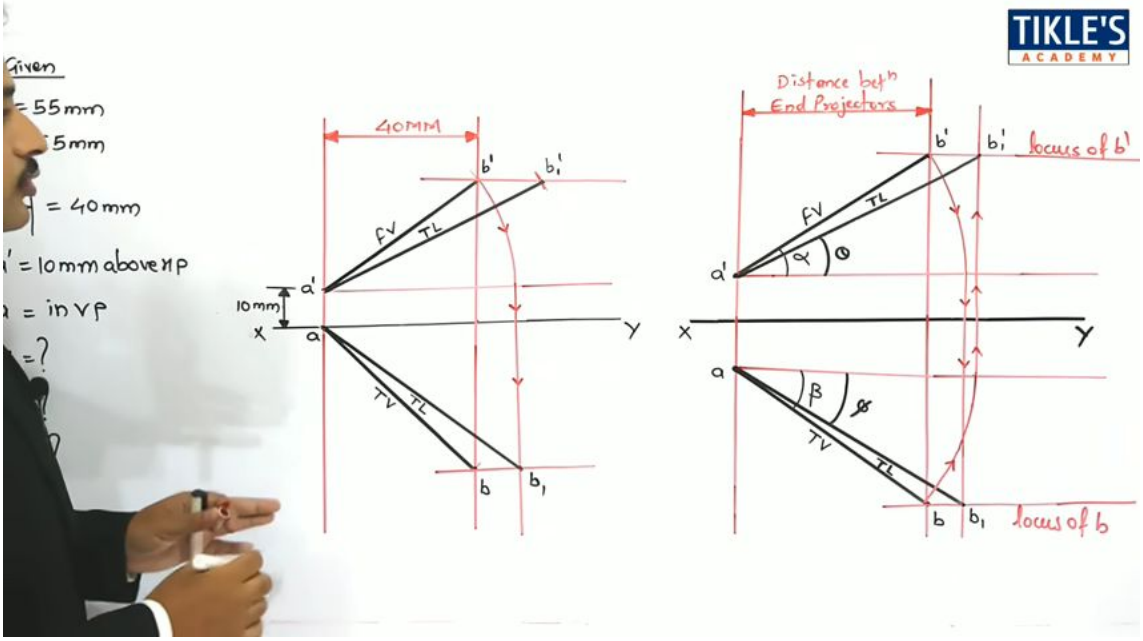
528.388069 Now, to draw FV of 55 mm, open the compass to 55mm, keep it at a' and draw an arc on the right hand end projector. Then, open for 65mm for TV, keep at a and draw arc below XY.



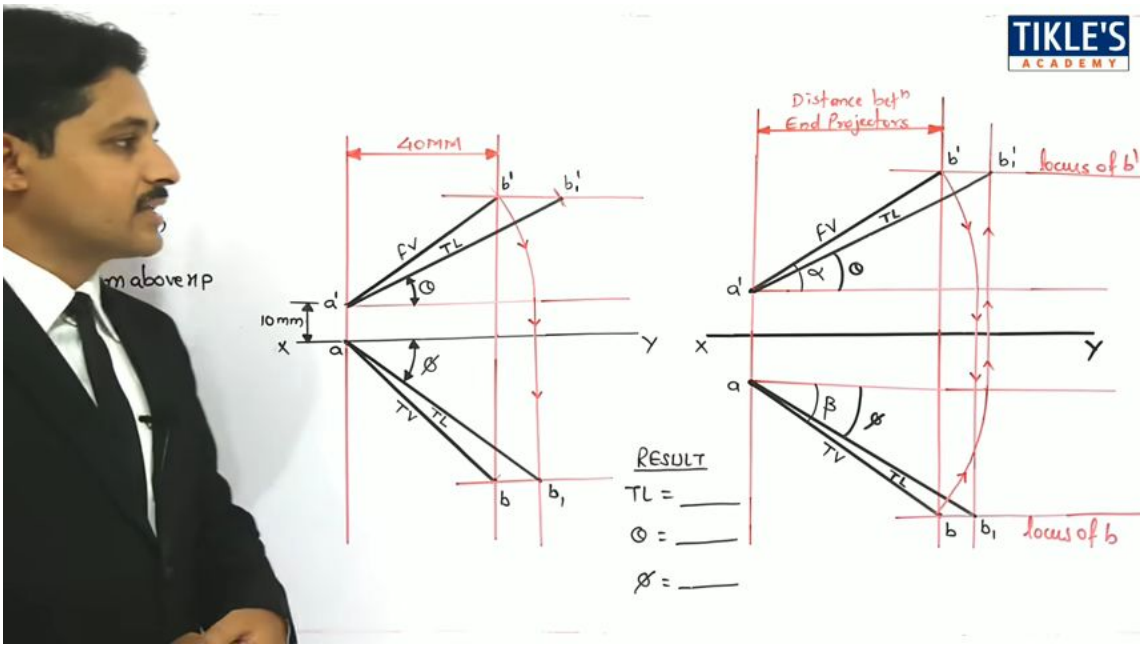
789.235424 now, for true length, open compass from a' to b', draw an arc till horizontal line through a' then extend that arc as a vertical line. this line's intersection at a horizontal line via b gives b1. Join b1 and a for true length.



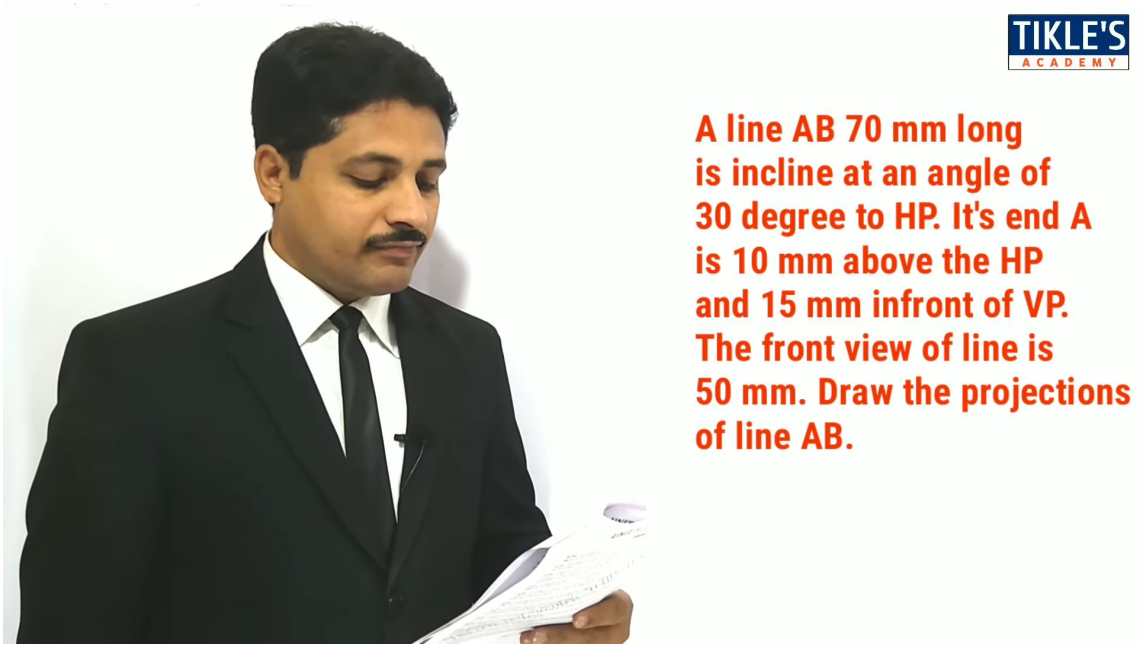
837.46713 now, true length is same so just draw the locus (horizontal line via b') and open compass along ab1 to measure TL and cut an arc on b' locus TAKING a' AS A CENTRE.



878.69334 label theta and phi. Write TL and both the angles in bottom right corner as result.



Q2. 67.790167



Length of line = TRUE LENGTH.